

Angular Blocks for the Compact Shell Proxy

Round 2 analytic note

Abstract

The compact shell proxy can be reduced to an angular rounding defect $\mathcal{E}_{\text{ang}}$ and a smooth positive payment $\mathcal{H}$. This note attacks the remaining defect. First, an exact Stieltjes formula is given with the strict common-jump convention made explicit. The negative mean of every complete angular cell is combined with the single-peaked shape of $q = -A'$ to obtain continuous block estimates. A paired radial block argument then controls the full shifted sample sum, including all common jumps, by

$$\mathcal{E}_{\text{ang}} < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2},$$

where N is the number of active shifted radial layers. On the actual compact residual $N \geq 1$. The purely analytic replacement is therefore reduced to one smooth scalar inequality, displayed in (28). That scalar inequality is not proved here.

1 Definitions and the target payment

For $a > 0$ and $z \geq 0$, define

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix $0 < \rho < 1$ and $K > 0$, and put

$$A(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1 - \rho)K}{\pi}. \quad (2)$$

The function A decreases strictly from T to 0. Denote its decreasing inverse by $R : [0, T] \rightarrow [0, K]$, and set

$$F(t) := R(t)^2, \quad q(z) := -A'(z), \quad \tau := A(\rho K) = KG_1(\rho). \quad (3)$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (4)$$

Consequently

$$0 \leq q(z) \leq \frac{1}{2}, \quad q \text{ increases on } (0, \rho K), \quad q \text{ decreases on } (\rho K, K). \quad (5)$$

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}, \quad N := \#\{n \geq 1 : x_n < T\}. \quad (6)$$

For $1 \leq n \leq N$, write

$$r_n := R(x_n), \quad m_n := M(r_n), \quad (7)$$

and define the exact angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{n=1}^N (m_n^2 - r_n^2). \quad (8)$$

The strict convention is built into M : at $r = m + 1/2$ one has $M(r) = m$, not $m + 1$.

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned} \mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \end{aligned} \quad (9)$$

The preceding structural reduction proves

$$\mathcal{E}_{\text{ang}} < \mathcal{H} \implies P(\rho, K) < W(\rho, K). \quad (10)$$

This note derives a smooth upper bound for the left side.

2 Complete angular cells and the single peak

Define the angular sawtooth error

$$e(r) := M(r)^2 - r^2. \quad (11)$$

For $m \geq 1$, the strict m -cell is

$$I_m = (m - 1/2, m + 1/2], \quad e(r) = m^2 - r^2 \quad (r \in I_m).$$

Endpoint values do not affect the following Lebesgue integrals. The positive and negative half-cell masses are

$$P_m := \int_{m-1/2}^m (m^2 - r^2) dr = \frac{m}{4} - \frac{1}{24}, \quad (12)$$

$$Q_m := - \int_m^{m+1/2} (m^2 - r^2) dr = \frac{m}{4} + \frac{1}{24}. \quad (13)$$

In particular,

$$\int_{m-1/2}^{m+1/2} e(r) dr = P_m - Q_m = -\frac{1}{12}. \quad (14)$$

Lemma 1 (Weighted complete-cell bounds). *Let $m \geq 1$ and suppose the complete cell $[m - 1/2, m + 1/2]$ lies in $[0, K]$.*

1. If $m + 1/2 \leq \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq -\frac{q(m)}{12} \leq 0. \quad (15)$$

2. If $m - 1/2 \geq \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq P_m q(m - 1/2) - Q_m q(m + 1/2). \quad (16)$$

3. On the unique cell, if any, whose interior contains ρK ,

$$\int_{m-1/2}^{m+1/2} e(r)q(r) dr \leq P_m q(\rho K). \quad (17)$$

The initial partial cell $0 \leq r \leq 1/2$ contributes nonpositively.

Proof. On the left half of a cell $e \geq 0$, and on the right half $e \leq 0$. If q is increasing, then $q(r) \leq q(m)$ on the positive half and $q(r) \geq q(m)$ on the negative half. Equations (12)–(14) give (15).

If q is decreasing, then $q(r) \leq q(m - 1/2)$ on the positive half and $q(r) \geq q(m + 1/2)$ on the negative half. This gives (16). On the peak cell, discard the negative half and use $q \leq q(\rho K)$ on the positive half. Finally, $M(r) = 0$ and $e(r) = -r^2$ for $0 \leq r \leq 1/2$. $\square$

Thus the continuous angular defect has a favorable analytic block structure:

$$\int_0^T e(R(t)) dt = \int_0^K e(r)q(r) dr. \quad (18)$$

All complete cells below the peak are nonpositive, and the decreasing side is controlled by the endpoint differences in (16). The difficulty is passing from the integral in (18) to the shifted samples in (8); the next subsection records that passage with all common jumps included.

3 Exact common-jump convention

Let

$$h(t) := e(R(t)), \quad C(t) := \#\{n : 0 < x_n \leq t, x_n < T\}, \quad p(t) := C(t) - t. \quad (19)$$

Choose the right-continuous versions of C, p , and h . Right continuity of h is exactly the strict angular convention: when $R(t) = m + 1/2$, the post-jump value uses $M = m$. The atomic measure dC therefore samples the correct value even when a radial point x_n is also an angular wall.

Lemma 2 (Euler identity with common jumps). *With the preceding representatives,*

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t) dt - \int_{(0,T]} p(t^-) dh(t). \quad (20)$$

At a common radial-angular wall the second integral uses the left value $p(t^-) = -3/4$; no extra or omitted wall term is permitted.

Proof. The measure dC has a unit atom at every active x_n , so

$$\mathcal{E}_{\text{ang}} = \int_{(0,T)} h(t) dC(t).$$

Since $dC = dt + dp$,

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t) dt + \int_{(0,T]} h(t) dp(t).$$

For right-continuous functions of bounded variation, the product formula in the form

$$\int_{(0,T]} h(t) dp(t) = [hp]_0^T - \int_{(0,T]} p(t^-) dh(t)$$

already contains the product of simultaneous jumps. Here $p(0) = 0$ and $h(T) = e(0) = 0$, so the boundary term vanishes. This proves (20). If $t = x_n$, then $C(t^-) = n - 1$ and hence $p(t^-) = n - 1 - (n - 1/4) = -3/4$, including a common angular wall. $\square$

Equation (20) explains why the negative means in (14) cannot be used without a jump audit. The second term contains both the continuous rise inside each angular cell and the downward atom at its strict lower wall. A direct sign deletion would be invalid. The paired estimate below controls the integral and all its atoms simultaneously.

4 A paired radial-block estimate

The cap $q \leq 1/2$ in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift $x_n = n - 1/4$ leaves a left block of length $3/4$, enough to pay the worst angular rounding in that layer.

Lemma 3 (One-layer paired block). *For every active n ,*

$$r_n \leq \frac{4}{3} \int_{n-1}^{x_n} R(t) dt - \frac{3}{4}, \quad (21)$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \quad (22)$$

Both statements remain valid, with the displayed strictness, at every radial-angular common wall.

Proof. Fix $t \in [n - 1, x_n]$ and put $z = R(t) \geq r_n$. Since $A(r_n) = x_n$, (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^z q(s) ds \leq \frac{z - r_n}{2}.$$

Hence

$$R(t) \geq r_n + 2(x_n - t).$$

Integration over the interval of length $3/4$ yields

$$\int_{n-1}^{x_n} R(t) dt \geq \frac{3}{4} r_n + \frac{9}{16},$$

which is (21).

By strict counting, $m_n < r_n + 1/2$ even when r_n is a half-integer: at $r_n = m + 1/2$ the value is $m_n = m$. Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (22). $\square$

Proposition 4 (Global angular block bound). *For every $0 < \rho < 1$ and every $K > 0$ with $N \geq 1$,*

$$\boxed{\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{N}{2}}. \quad (23)$$

In particular,

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1 - \rho^2)K^2}{6} - \frac{1}{2}. \quad (24)$$

Proof. The left blocks $[n - 1, x_n]$, $1 \leq n \leq N$, are disjoint subsets of $[0, T]$. Summing (21) and then (22) gives

$$\begin{aligned} \mathcal{E}_{\text{ang}} &< \sum_{n=1}^N \left(r_n + \frac{1}{4} \right) \\ &\leq \frac{4}{3} \sum_{n=1}^N \int_{n-1}^{x_n} R(t) dt - \frac{N}{2} \\ &\leq \frac{4}{3} \int_0^T R(t) dt - \frac{N}{2}. \end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t) dt = \int_0^K A(z) dz.$$

Scaling $z = as$ in (1) and using

$$\int_0^1 \sqrt{1 - s^2} ds = \frac{\pi}{4}, \quad \int_0^1 s \arccos s ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z) dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t) dt = \frac{(1 - \rho^2)K^2}{8},$$

which proves (23). Equation (24) follows from $N \geq 1$. □

5 The remaining scalar inequality

Set

$$\rho_c := \frac{1}{1 + 2\pi}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1 - \rho)^2} + 132}, \quad (25)$$

and consider the closed superset of the actual compact residual

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right\}. \quad (26)$$

On this set,

$$T \geq \frac{(1 - \rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1 - \rho)^2}{\pi^2}} > 1, \quad (27)$$

so $N \geq 1$. Proposition 4 and (10) reduce the analytic compact closure to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1 - \rho^2)K^2}{6} + \frac{1}{2} > 0} \quad ((\rho, K) \in \mathcal{R}_c). \quad (28)$$

Everything in $\mathcal{S}$ is smooth on a neighborhood of $\mathcal{R}_c$; the floor functions and all radial–angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned} \mathcal{S}(\rho, K) = & \frac{(5\rho^2 - 2)K^2}{12} - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1 + A(z))}{(1 + 2A(z))^2} dz. \end{aligned} \quad (29)$$

The elementary identity

$$\frac{u(1 + u)}{(1 + 2u)^2} = \frac{1}{4} - \frac{1}{4(1 + 2u)^2} \quad (30)$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned} \mathcal{S}(\rho, K) = & \frac{(1 + 2\rho^2)K^2}{12} - \frac{1}{2} \int_{\rho K}^K \frac{z}{(1 + 2A(z))^2} dz \\ & - \frac{\pi\rho K}{4(1 - \rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32} \right). \end{aligned} \quad (31)$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (31).

Remark 5 (Status and next step). The complete-cell bounds, the common-jump formula, and the global estimate (23) are proved analytically. The first remaining step in this route is the smooth scalar inequality (28). Proving it will require a uniform lower bound for the positive outer-action integral in (29), or, equivalently, a uniform upper bound for the boundary-layer loss in (31), together with the explicit endpoint terms. Until that inequality is established, this note does not replace the compact interval certificate.